

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE****Regular/Supplementary Winter Examination – 2024****Course: B.Tech****Branch : Electronics and Computer Engineering/ Electronics and****Computer Science****Semester : V****Subject Code & Name: BTECPC501 Computer Networks and Cloud Computing****Max Marks: 60****Date:06/02/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                                                         | (Level/CO) | Marks     |
|-------------|---------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1</b> | <b>Objective type questions. (Compulsory Question)</b>                                                  |            | <b>12</b> |
| 1           | <b>Identify the layer which provides service to the user.</b>                                           | Remember   | 1         |
|             | a. Session Layer    b. Physical Layer    c. Application Layer    d. Presentation layer                  |            |           |
| 2           | <b>What is the length of the MAC address?</b>                                                           | Remember   | 1         |
|             | a. 8 bits    b. 16 bits    c. 32 bits    d. 48 bits                                                     |            |           |
| 3           | <b>What is the collection of the hyperlinked document on the internet known as?</b>                     | Remember   | 1         |
|             | a. HTML    b. Email    c. www    d. Internet                                                            |            |           |
| 4           | <b>Which of the following layer isn't present in the TCP/IP model but is included in the OSI model?</b> | Remember   | 1         |
|             | a. Network layer    b. Session layer    c. Application layer    d. Transport layer                      |            |           |
| 5           | <b>Identify among the following which belongs to class A.</b>                                           | Remember   | 1         |
|             | a. 121.12.12.248    b. 128.12.12.248    c. 129.12.12.248    d. 130.12.12.248                            |            |           |
| 6           | <b>Identify the switching method in which the message is divided into small packets.</b>                | Remember   | 1         |
|             | a. Virtual switching    b. Packet switching    c. Message switching    d. None                          |            |           |
| 7           | <b>Identify the total versions of IP.</b>                                                               | Remember   | 1         |
|             | a. 1    b. 2    c. 3    d. 4                                                                            |            |           |
| 8           | <b>The arrangement where all data pass through a central computer is known as</b>                       | Remember   | 1         |
|             | a. Ring topology    b. Mesh topology    c. Star topology    d. Bus topology                             |            |           |

|             |                                                                                                                     |                                  |                                              |                      |            |           |
|-------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------|----------------------|------------|-----------|
| 9           | <b>Choose among the following, which are the characteristics of IaaS.</b>                                           |                                  |                                              |                      | Remember   | 1         |
|             | a. Highly scalable                                                                                                  | b. Flexible and dynamic platform | c. Facilitates hardware access to many users | d. All of mentioned  |            |           |
| 10          | <b>Choose the most important use of cloud computing.</b>                                                            |                                  |                                              |                      | Remember   | 1         |
|             | a. Data privacy                                                                                                     | b. Data storage                  | c. Cost-effective                            | d. Security          |            |           |
| 11          | <b>Identify the deployment model among the following.</b>                                                           |                                  |                                              |                      | Remember   | 1         |
|             | a. Private                                                                                                          | b. Public                        | c. Hybrid                                    | d. All of mentioned  |            |           |
| 12          | <b>From which layer does the data link layer collect packets and encapsulate them into frames for transmission?</b> |                                  |                                              |                      | Remember   | 1         |
|             | a. Transport layer                                                                                                  | b. Physical layer                | c. Network layer                             | d. Application layer |            |           |
|             |                                                                                                                     |                                  |                                              |                      |            |           |
| <b>Q. 2</b> | <b>Solve the following.</b>                                                                                         |                                  |                                              |                      |            | <b>12</b> |
| A)          | Explain the different network topologies.                                                                           |                                  |                                              |                      | Understand | <b>6</b>  |
| B)          | Compare OSI reference model and TCP/IP                                                                              |                                  |                                              |                      | Understand | <b>6</b>  |
|             |                                                                                                                     |                                  |                                              |                      |            |           |
| <b>Q.3</b>  | <b>Solve the following.</b>                                                                                         |                                  |                                              |                      |            | <b>12</b> |
| A)          | What are the data link layer design issues? Discuss.                                                                |                                  |                                              |                      | Understand | <b>6</b>  |
| B)          | How CRC is used to detect error? Explain with an example.                                                           |                                  |                                              |                      | Apply      | <b>6</b>  |
|             |                                                                                                                     |                                  |                                              |                      |            |           |
| <b>Q. 4</b> | <b>Solve Any Two of the following.</b>                                                                              |                                  |                                              |                      |            | <b>12</b> |
| A)          | What is DNS? Explain in detail.                                                                                     |                                  |                                              |                      | Understand | <b>6</b>  |
| B)          | Explain go-back algorithm for flow control.                                                                         |                                  |                                              |                      | Understand | <b>6</b>  |
| C)          | What is cloud computing? Give advantages and disadvantages of cloud computing                                       |                                  |                                              |                      | Apply      | <b>6</b>  |
|             |                                                                                                                     |                                  |                                              |                      |            |           |
| <b>Q.5</b>  | <b>Solve Any Two of the following.</b>                                                                              |                                  |                                              |                      |            | <b>12</b> |
| A)          | Explain cloud computing deployment models.                                                                          |                                  |                                              |                      | Understand | <b>6</b>  |
| B)          | Explain the concept of electronic mail and the world wide web.                                                      |                                  |                                              |                      | Understand | <b>6</b>  |
| C)          | State the different routing algorithms and describe shortest path algorithm in detail                               |                                  |                                              |                      | Understand | <b>6</b>  |
|             |                                                                                                                     |                                  |                                              |                      |            |           |
| <b>Q. 6</b> | <b>Solve Any Two of the following.</b>                                                                              |                                  |                                              |                      |            | <b>12</b> |
| A)          | Explain IaaS, SaaS and PaaS services in cloud computing                                                             |                                  |                                              |                      | Understand | <b>6</b>  |
| B)          | Describe the classes of IPV4 addressing                                                                             |                                  |                                              |                      | Understand | <b>6</b>  |
| C)          | Explain the concept of data center in cloud computing.                                                              |                                  |                                              |                      | Understand | <b>6</b>  |
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